Resume

Bui Xuan Toan

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Educational Record

- **2007–2009:** Le Quy Don High School, Phan Rang – Thap Cham City, Ninh Thuan Province
- **2009–2013:** Bachelor's Degree in Finance-Banking, Ho Chi Minh City Open University
- **2015–2017:** Master's Degree in Industrial Management, Graduate Institute of Industrial Management, National Central University, Taiwan

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Work Experience

- **2013–2015:** Financial Analysis – Creditor (Personal Customer) at VietinBank Ninh Thuan, Ninh Thuan Province
- **07/2017–08/2018:** Planning Function Developer at Fusheng Precision, Taoyuan, Taiwan
- **08/2018–10/2020:** Business Process Automation at Grab (Regional Team)
- **11/2020–08/2021:** Data Engineer at Just Analytics
- **09/2021–12/2023:** Data Engineer (Project Lead) at HCL
- **01/2024–06/2024:** Data Engineer / Solution Architect (Independent Contractor) at HailstoneLab
- **07/2024–02/2025:** Senior Data Engineer (Contractor) at Grab (Data Ingestion Team)
- **02/2025–present:** Senior Data Engineer at Persefoni

Certificates

- TOEIC: 805 (2011)
- IELTS: Listening 8.0 | Reading 7.0 | Writing 6.0 | Speaking 6.0 | Overall 7.0 (2014)
- Google Cloud Data Engineer (2019)
- Google Cloud Data Architect (2020)
- Google Cloud DevOps Engineer (2023)
- Google Cloud Security Engineer (2023)
- AWS Certified DevOps Engineer – Professional (2024)
- AWS Certified Solutions Architect – Professional (2024)
- Databricks Certified Data Engineer Professional (2024)

- SnowPro Core Certification (2025)

Skills

Core CS & Software Engineering

- Solid understanding of computer science concepts, statistics, and software development
- Data structures, classic algorithms, and complexity analysis
- Linux administration, containerization (Docker), container orchestration (Kubernetes)
- Process, thread, and async programming
- HTTP protocol, REST API design
- Strong advocate of **software quality** — apply **Test-Driven Development (TDD)** as a core practice: write tests first, drive design through failing tests, and refactor with confidence
- Practice **Continuous Integration / Continuous Deployment (CI/CD)**: automate build, test, and deployment pipelines to catch issues early and ship reliably
- Apply **Clean Code principles** (Robert C. Martin) daily: meaningful naming, small focused functions, single responsibility, minimal side effects, and keeping code readable as a first-class concern
- Apply **Clean Architecture** to separate business logic from infrastructure, making systems easier to test, maintain, and evolve

Python

- Fluent in Python with strong knowledge of built-in data structures
- Object-oriented design and implementation
- Support libraries for cloud services (AWS, GCP)
- Performance improvement using `asyncio` and `httpx`
- Test-driven development and code quality focus

Data Analytics

- Data wrangling with pandas, numpy, and Jupyter Notebook
- Data visualization with matplotlib, seaborn, and plotly
- Data preparation and cleaning for optimization models

Data Engineering & Data Warehousing

- Advanced SQL query writing and optimization across relational databases
- Experience with traditional databases: PostgreSQL, MySQL, SQL Server
- Experience with columnar/cloud databases: BigQuery, Presto (Trino), Databricks Delta Tables, Snowflake
- Distributed data processing: Apache Spark, Apache Beam, Apache Flink
- **Data warehouse design** focusing on scalable architectures (Star/Snowflake schema, Medallion architecture)
- **Data Reliability & Testing**: Implement robust testing frameworks (dbt tests, custom validations) to ensure **data correctness** and **freshness**; apply TDD to data modeling
- **Observability**: Implement end-to-end monitoring for data platforms, tracking pipeline health, data quality, and lineage; **leverage query tagging** (e.g., Snowflake query tags) to perform granular analysis of query execution time and cost across complex data models
- **Cost Optimization**: Optimize cloud warehouse spend (Snowflake, BigQuery) through query tuning, resource management, and storage optimization (e.g., achieved 33% cost reduction)
- Orchestration with Apache Airflow
- Change Data Capture (CDC) with Debezium, Kafka Connect, Kafka Server
- Kafka producer/consumer using confluent-kafka Python client

- Data warehouse design and testing with dbt (test-driven approach)

Visualization

- Apache Superset, Looker Studio, Tableau, Power BI
- CI/CD design for visualization platforms (version control, safe deployment)
- Cost optimization with caching and logical data model tuning
- Multi-tenant access control setup

Backend & AI

- REST API development with Flask, Django, FastAPI
- LLM agent development using LangChain, LangGraph, LangServe

Vibe Coding (AI-Assisted Software Development)

- Proficient in AI-assisted development using tools such as Claude Code, GitHub Copilot, and Cursor to accelerate feature delivery without sacrificing quality
- Apply **Test-Driven Development (TDD)** as the anchor of AI-assisted workflows: write failing tests first, use AI to generate implementation, then refactor — ensuring generated code is always validated against a clear specification
- Treat AI-generated code with the same rigor as human-written code: review for correctness, security, edge cases, and maintainability before accepting
- Use AI tools to scaffold boilerplate, suggest refactors, and explore approaches, while retaining full ownership of design decisions and code quality
- Enforce code quality gates (linting, type checking, test coverage thresholds) in CI/CD pipelines to catch issues in AI-generated code automatically
- Focus on prompt engineering for code generation: provide context, constraints, and expected behavior to guide AI toward correct and minimal solutions

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Research Experience

- Applied Genetic Algorithm to Schedule Brewery Production *(Master's thesis)*
- Breast Cancer Prediction *(Machine Learning coursework)*
- American Presidential Election Prediction *(NLP coursework – correctly predicted Trump's win)*

Highlight Projects

Project	Company	Scope	Result
Compute WIP	Fusheng Precision	Fusheng Vietnam	Reduced WIP report generation time from hours to seconds
Highlight Late Kanban Cards	Fusheng Precision	Fusheng Vietnam	Helped managers identify late cards for further investigation
Compute Incentive for Grab Express	Grab	Grab Vietnam	Replaced manual Excel computation. Stack: Python, SQL, pandas, Athena, Google API
Automation Send Email for Drivers' Incidents	Grab	Grab Vietnam	Eliminated manual email writing. Stack: Python, SQL, pandas, Athena, AWS SES

Data Analysis for QC Team	Grab	Grab Vietnam	Cleaned database for better driver management. Stack: Python, MySQL, pandas, Athena, Google API
Collect Drivers' Commission Fee for Partners	Grab	Grab Indonesia	Faster and more accurate calculations. Stack: Python, MySQL, pandas, Athena, Google API
Grab Hitch	Grab	Grab Hitch Singapore	Supported Hitch team's core calculations (points, tier, commission) for 100K+ users. Stack: Python, MySQL, pandas, Athena, Google API
Mex Correction	Grab	Grab Food Regional	Summarized driver restaurant incident reports to reduce ticket volume. Stack: Python, MySQL, pandas, Athena, Google API
Mex Insight	Grab	Grab Food Regional	Automated delivery report process for restaurant partners. Stack: Python, MySQL, pandas, Athena, Google API
Mex Billing	Grab	Grab Food Regional	Built billing system for Grab Food. Stack: Python, MySQL, pandas, Athena, Google API
Dax Event Tracker	Grab	Grab Regional	Automated event invitations, registration, and analysis. Stack: Python, MySQL, pandas, Athena, Google API
Data Lake	Just Analytics	Nam Long Corporation	Custom data pipeline for finance and assets data. Stack: Python, SQL, SQL Server
Data Lake Migration	Just Analytics	Grab	Migrated data pipeline from AWS to Azure. Stack: Python, SQL, Azure Databricks
KOL Database	HCL	Amorepacific	Deployed Airflow with Docker, built pipeline for YouTube channel/video data. Stack: Python, SQL, web scraping, AWS RDS MySQL
Mall Review Database	HCL	Amorepacific	Built web crawling pipeline for product reviews. Stack: Python, SQL, web scraping, AWS RDS MySQL
Maintain Batch Data Pipeline	HCL	ANZ	Automated metadata creation, query partition optimization. Stack: Python, SQL, BigQuery, Airflow, dbt
Group Management System	HCL	ANZ	GCP IAM-based group management. Stack: Git, Terraform, Bash

Deploy Datahub to Shared Cluster	HCL	ANZ	Deployed Datahub to GKE shared cluster. Stack: Helm, Kubernetes, Bash
Build em_helper Python Package	HCL	ANZ	Support package for Vault and Kafka communication. Stack: Python, Kafka
Infrastructure as Code	Contractor	HailstoneLabs	Centralized GCP resource provisioning for dev and prod. Stack: GitHub, Terraform, Cloud Build
CI/CD for Airflow	Contractor	HailstoneLabs	Built CI/CD to build and deploy Airflow. Stack: Helm, GKE, GCP Artifact Registry, Cloud Build, Docker
Role-Based Access Management	Contractor	HailstoneLabs	Implemented and transferred knowledge on RBAC
Twitter Dashboard Data Pipeline	Contractor	HailstoneLabs	Built pipeline to track VC activity on Twitter. Stack: Python, SQL, Airflow, web scraping, dbt, BigQuery. Saved ~$40,000/month by replacing Enterprise API with web scraping
Crypto Market Tracking Pipeline	Contractor	HailstoneLabs	Dashboard tracking crypto market movements. Stack: Python, SQL, Airflow, dbt, BigQuery, Slack app
ETH Block Data Pipeline	Contractor	HailstoneLabs	Ingested Ethereum node data, optimized for queries. Stack: Python, SQL, Airflow, dbt, BigQuery
Hugo LLM Agents for Auto-Healing	Contractor	Grab	Built and deployed LLM agents for automated data pipeline debugging. Stack: LangChain, LangGraph, LangServe
Airflow 1 → Airflow 2 Migration	Contractor	Grab	Migrated pipelines to modern Python-based Airflow with improved secrets management
Cross-Region Data Copy (SG → VN)	Contractor	Grab	Copied large volumes of data from Singapore to Vietnam AWS for Decree 53 compliance
OVO Data Warehouse Migration (GCP → AWS)	Contractor	Grab	Migrated OVO Indonesia data pipelines and warehouse from GCP to AWS
Scope 1/2/3 Emission Data Model	Persefoni	Persefoni	Fixed long-standing bugs causing incorrect calculations; applied TDD for stable releases. Stack: Debezium, Kafka, Snowflake, dbt, GoodData
Waste & Water Data Pipeline	Persefoni	Persefoni	Delivered bug-free data model and visualization through thorough testing

Financial Data Model (France & Other Markets)	Persefoni	Persefoni	Extended model for multi-language/market support; fixed hidden bugs
Databricks ETL → CDC Migration	Persefoni	Persefoni	Improved data freshness using CDC and Snowflake streaming architecture
Snowflake Cost Optimization	Persefoni	Persefoni	Reduced Snowflake costs by ~33%
Emission & Energy Use Intensity Model	Persefoni	Persefoni	Built model supporting facility occupied area changes over time and complex intensity metric computation